

Deep Latent Variable Models for Text and Audiovisual Data

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Latent Variables: New Data, New Challenges

- Many latent variable models in the social sciences (e.g., ideal point models based on voting records, latent factors in SVAR models from macro time-series).
 - Unstructured data is plentiful and offers new research opportunities, e.g. texts, images, audio, video.
 - BUT unstructured data often means:
 - Large number of observations (*large n*)
 - High dimensionality (*large p*)
 - Multiple modalities (e.g., *text and images and videos*)
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- **Deep latent variable models**

- Fairly general class of latent variable models that nests many special cases
- Extended regularized autoencoder to approximate the marginal likelihood
- Fast, scalable, multimodal, allows for covariates and auxiliary outcomes
- Good finite-sample performance in Monte Carlo simulations

- **Applications in topic modeling for social science:**

- Introduce the Generalized Topic Model (GTM).
- Unpack speech polarization in the US Congress.
- Nowcast macroeconomic outcomes with business news.
- Jointly analyze text and images of posts by U.S. politicians on Facebook.

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Structural Model

- Data:
 - W_m = vector of document response variables for each modality m (e.g. text embeddings).
 - Y = vector of auxiliary outcomes (e.g. economic growth).
 - X^p , X^c , and X^s = covariates influencing the latent, response, or outcome variables, respectively (e.g. political party of speaker).
- Z = vector of latent variables (e.g. topics to be learned).
- We assume that

$$Z \sim P_{Z|X^p} \quad \text{and} \quad W_m = G_m(Z, X^c) \quad \text{and} \quad Y = F(Z, X^s).$$

where P_Z is prior distribution on Z , specified as GLM.

→ Fairly general formulation, as P_Z , $G_1, \dots, G_m$, and F are left unspecified (we will parametrize them later for estimation).

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Estimation: Overview

For response variables $\{w_m\} \in \{1, \dots, M\}$, outcome y , latent variables z and covariates x^p , x^c , and x^s :

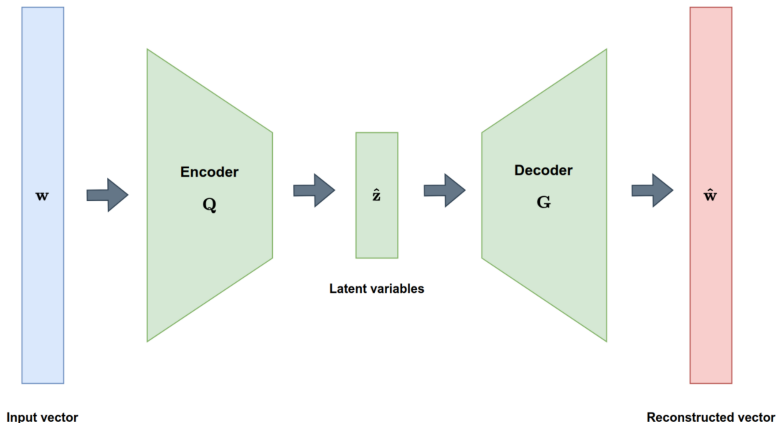
$$P(w_1, \dots, w_m, y \mid x^p, x^c, x^s) = \int_{\mathcal{Z}} P(z \mid x^p) P(y \mid z, x^s) \prod_{m=1}^M P(w_m \mid z, x^c) dz.$$

→ **Problem:** The marginal likelihood is *intractable*.

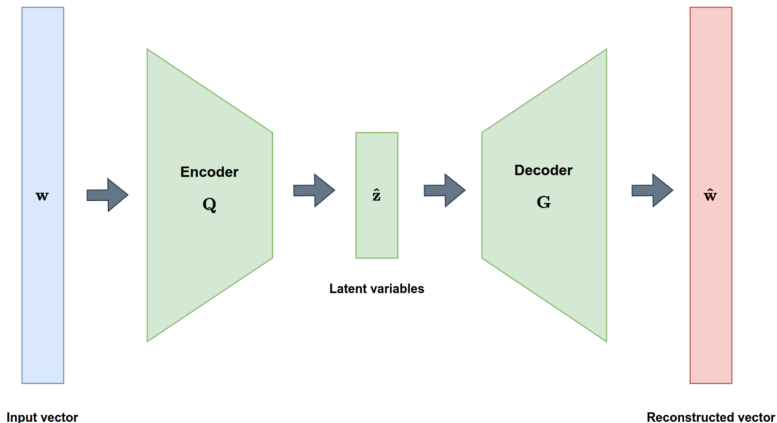
→ **Solution:** Approximate $P(w_1, \dots, w_m \mid x^p, x^c)$ as a regularized variational autoencoder (VAE):

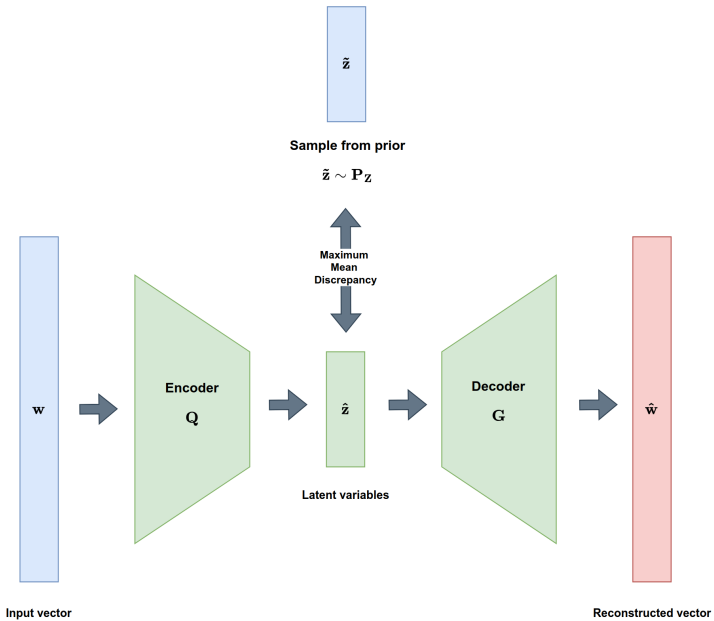
- Encoder $Q(w_1, \dots, w_m, x^p) \approx P(z \mid w_1, \dots, w_m, x^p)$
- Decoders $G_m(z, x^c) \approx P(w_m \mid z, x^c)$
- Predictor $F(z, x^s) \approx P(y \mid z, x^s)$
- Nudge $Q(w_1, \dots, w_m, x^p)$ to remain close to the prior distribution.

- **Encoder** is a feed-forward neural net mapping high-dimensional inputs w (e.g. text) to low-dimensional $\hat{z}$ (encodings / embeddings / topics).
- **Decoder** is a feed-forward neural net mapping low-dimensional $\hat{z}$ back to high-dimensional $\hat{w}$ (reconstructed input).



- **Input vector w** can be any “document” representation; e.g. bag-of-words/phrases, or LLM/VLM embeddings.
- **Reconstruction vector w** can be different from input – using bag-of-words is recommended because it produces interpretable topics.



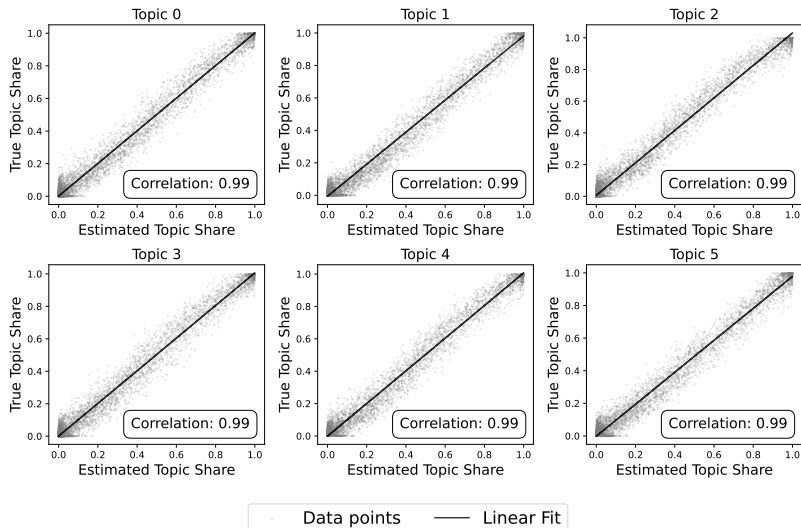


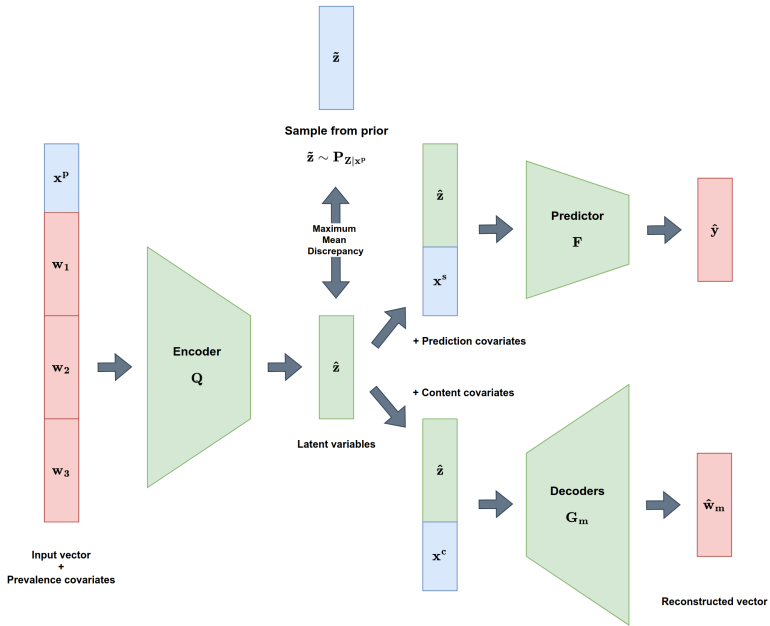
$$\text{TotalLoss} = \text{ReconstructionLoss}(w, \hat{w}) + \lambda \widehat{\text{MMD}}_k^2(\hat{z}, \tilde{z})$$

Model Consistency

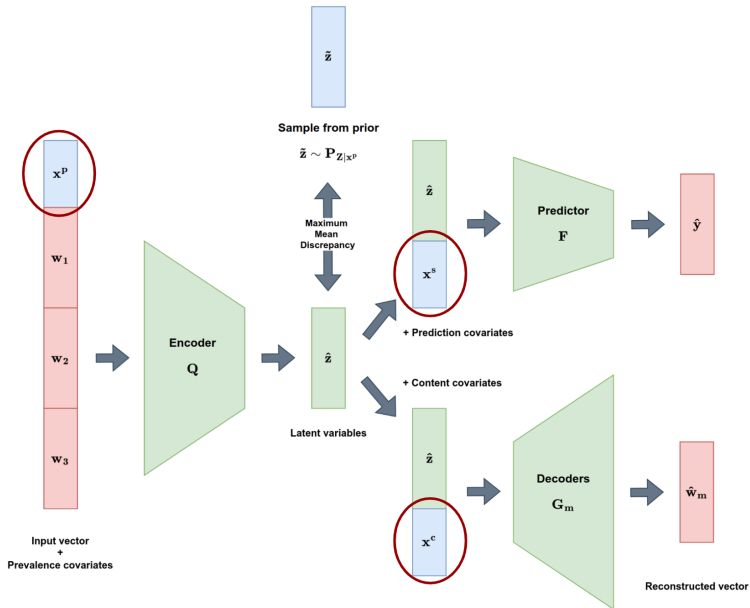
- Wang and Blei (2019): variational bayes consistently estimates posterior distribution parameters.
- Margossian and Blei (2024): amortized variational inference (SGD-like algorithm to train VAEs) is consistent for “simple hierarchical models” (Agrawal and Domke 2021), including VAEs.
- Also evaluate fit in simulations.

Monte Carlo Simulations for Topic Fit (Dirichlet Prior)





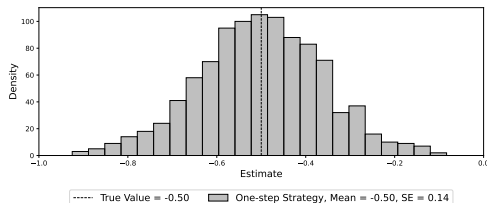
$$\text{TotalLoss} = \sum_{m=1}^M \text{RecLoss}(w_m, \hat{w}_m) + \lambda_0 \widehat{\text{MMD}}_k^2(\hat{z}, \tilde{z}) + \lambda_1 \text{PredLoss}(y, \hat{y})$$



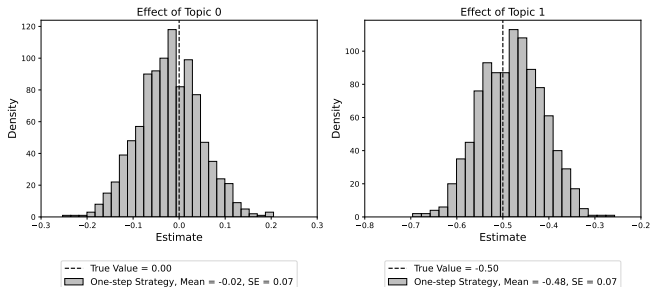
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Monte Carlo Simulations for Learning Effects of Covariates

Simulation: Density of Estimates for Effect of Covariates on Prior



Simulation: Density of Estimates for Effect of Topic Shares on Outcome



Analyzing Covariates: U.S. Congressional Record

- Estimate GTM on the full *US Congressional Record* (1873 – 2016).
- Options:
 - bag-of-n-grams representation with vocab of 14K n-grams (n=1,2,3)
 - 50 topics
 - Allow party and time covariates to affect topic choice and topic framing.
- Training is fast – less than an hour on GPU, compared to a few hours for LDA (Mallet), or a few days for STM (structural topic model).

Learned Topics with Party Associations

Democrat-Associated Topics

1. **Elections and Voting.** elector, ballot, vacancy, suffrage, legislative power, fill vacancy
2. **Legal Claims.** claimant, pay claim, claim claim, attorney fee, statute limitation, claim pay
3. **Foreign Aid.** foreign aid, military assistance, military aid, military budget, economic aid, aid
4. **Judiciary and Trials.** jury trial, jury, trial jury, grand jury, district judge, defendant
5. **Public Buildings and Construction.** circuit judge, cubic, public building, lieutenant, cadet, erection
6. **Energy Policy.** change exist law, free election, national energy policy, testimony hearing, change exist, debatable
7. **Law Enforcement.** law enforcement officer, enforcement officer, absent, pair, prisoner war, enlisted man
8. **Military Weapons.** missile, ballistic missile, ballistic, submarine, silo, cruiser
9. **Judicial and Highway Funding.** district attorney, highway trust fund, highway trust, judgeship, parking, controller
10. **War Negotiations.** military aid, aggression, embargo, peace process, troop, peace
11. **Banking Regulation.** hold company, national bank, banking, bank bank, bank, loan association
12. **Public Health.** home health, disease, heart disease, mental health, health center, dental
13. **Crime and Justice.** juvenile, crime, deport, firearm, violent crime, handgun
14. **Postal Services.** cent pound cent, pound cent, cent pound, postal employee, pound, advertising

Republican-Associated Topics

1. **Taxation.** sale tax, property tax, tax income, property taxes, tax return, local taxes
2. **Land Entitlement.** vessel, act entitle act, certain land, act entitle, subdivision, entitle act
3. **National Forests.** national forest, wilderness, acre, wilderness area, grazing, timber
4. **Water and Navigation.** navigable, nuclear waste, dam, flood control, basin, navigation
5. **Condolences and Tribute.** condolence, mourn, untimely death, eulogy, passing, sense humor
6. **Pensions and Social Security.** grant pension, pension, increase pension, widow, social security benefit, retirement system
7. **Budget and Appropriations.** budget authority outlay, authority outlay, budget authority, amount recommend
8. **Labor Market.** wage rate, minimum wage, increase minimum wage, wage worker, cent hour, farm labor
9. **Legislative Process.** unfinished business, divide control, general appropriation, unfinished, open hour, complete action
10. **Housing and Urban Development.** affordable housing, public housing, urban renewal, housing unit, housing, voucher
11. **Welfare and Unemployment.** food stamp, unemployment benefit, welfare reform, money borrow, poor people
12. **Transportation and Infrastructure.** passenger, airport, highway trust fund, highway trust, rail, mail service
13. **Forest Conservation.** forest, rainfall, pest, timber, insect, acre
14. **Military Service and Veterans.** active duty, veteran, disabled veteran, enlisted, reservist, enlisted man

Party-Specific Framing of Topics

Topic Label	Top Words
Welfare and Unemployment	R: money, man, pay , people, pension, come D: unemployment, poor people , poverty , hungry , food stamp, unemployed
Elections and Voting	R: precinct, election, contestant, elector, ballot, seat D: civil right , racial , equal protection , segregation , equal protection law, filibuster
Military Service and Veterans	R: officer , man, soldier, enlistment, regiment, enlisted man D: veteran, disabled veteran , personnel, care veteran , veteran family, veteran benefit
Campaign Finance	R: election, candidate, party, money, postage, expenditure D: campaign finance, spending limit, campaign finance reform , campaign contribution
Labor Market	R: laborer, labor, man, railroad, workman, railway D: minimum wage, wage worker, worker, increase minimum wage , cent hour, wage
Capital Gains and Taxes	R: taxation, cent, capital, taxis, pay, business D: capital gain, income, tax, tax cut , income tax, excise
Public Health	R: cord, care, service, hospital, physician , blood D: health, mental health, health service, heart disease, cancer, public health
Crime and Justice	R: officer , liquor, alien , consular, gun , intoxicate D: juvenile , law enforcement, drug, enforcement, crime, criminal justice

GTM Learns High-Quality Word Embeddings

Most Similar Phrases in Embeddings Space for:			
taxation	immigration	war	health
levy tax burden taxation taxes impose tax income tax power tax cent amount revenue taxes levy	immigration reform immigration law immigrant come illegal immigration deport immigrant undocumented deportation	military troop soldier war war country peace man nation	health care medical care disease cancer drug hospital child
wage	election	regulation	fiscal
minimum wage wage increase increase minimum wage increase minimum raise minimum wage worker labor wage hour	election hold ballot election election election law day election candidate general election voter	law regulate regulatory federal regulation propose regulation new regulation law regulation company	budget appropriation fund fiscal fiscal authorization increase expenditure fiscal budget

Notes. Most similar phrases to specific words in embeddings space, measured by cosine similarity. Embeddings correspond to 256-dimensional hidden layer in the decoder.

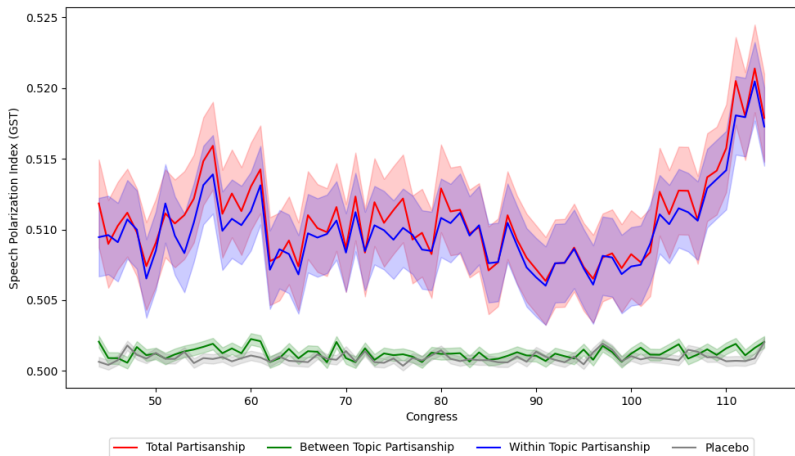
Decomposing Partisan Divergence in Speech

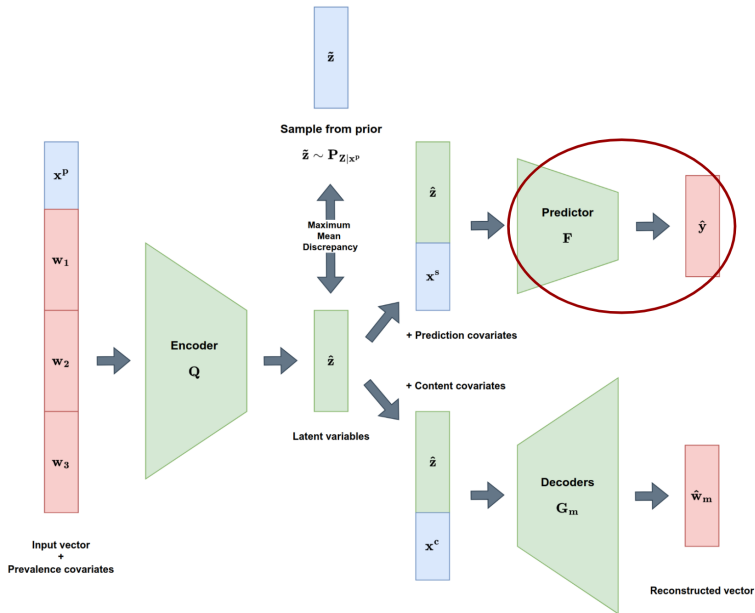
- Jensen et al (2012) and Gentzkow et al (2019) showed that U.S. congressional speech became more polarized starting in the 1990s.
- Was that due to topic choice or topic framing?
- Decomposition approach: Use fitted GTM (and altering x^p and x^c) to generate samples of partisan speeches over time for four counterfactual scenarios:
 - **Total polarization (observed):** Republicans and Democrats differ in topic choice and topical content.
 - **Between-topic polarization (topics):** Republicans and Democrats differ in topic choice but not in topical content.
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 - **Placebo:** Republicans and Democrats do not differ in either dimension.

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Slant matters (much) more than topic choice.





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Supervised-Learning Objective

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- The auxiliary learning objective, PredLoss, is a standard ML loss function, eg MSE or cross-entropy.
- nudges GTM to learn topics that are predictive of the auxiliary outcome Y .
- Set higher weights on λ_1 to improve prediction, at the potential cost of lower-quality topics.
- Here we illustrate prediction with macro forecasting.
- Another use case: semi-supervised topic model. If some documents have existing topics labeled, include those as Y multinomial labels and it will nudge GTM to use factors in Z to those topics.

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Supervised Learning: Macro Nowcasting

- Corpus: full text of *Wall Street Journal* articles, 1984-2012.
- Pre-processing: drop caps, digits, stopwords, one/two-letter words.
- Vocab: 14K words/bigrams filtered by document frequency.
- Prediction target: smoothed Log Industrial Production Growth in month after the article is published.
- GTM settings:
 - k=50 topics; encoder/decoder have one hidden layer of 256 dimensions with ReLU activation; predictor has no hidden layers and just linearly combines topics to predict the outcome.
 - 40/40/20 train/tune/test split, 20% dropout and early stopping.
- LDA baseline (Bybee et al 2023):
 - train vanilla LDA with 50 topics
 - fit lasso model on topics to predict same target.

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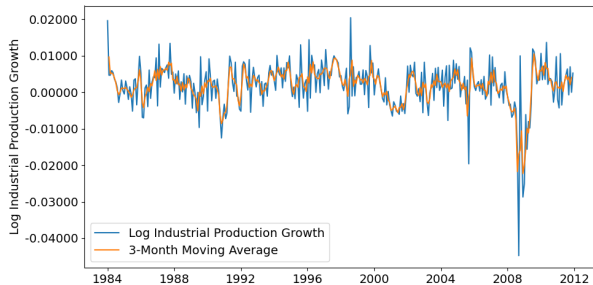
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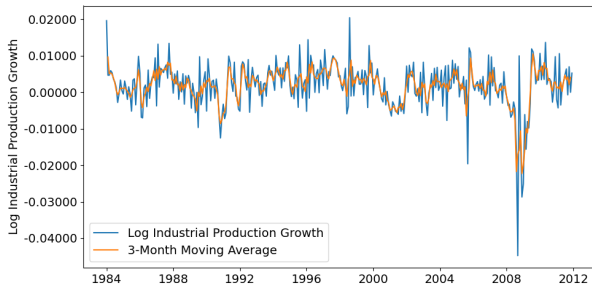
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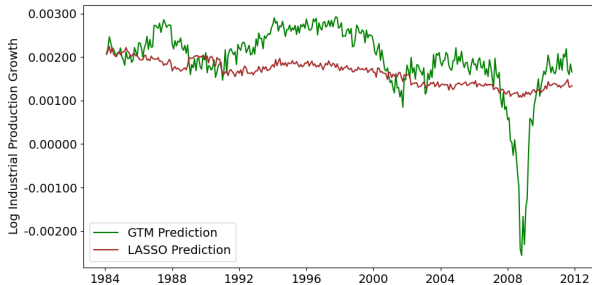


Predictions Based on Topics from the *Wall Street Journal*

Log Industrial Production Growth



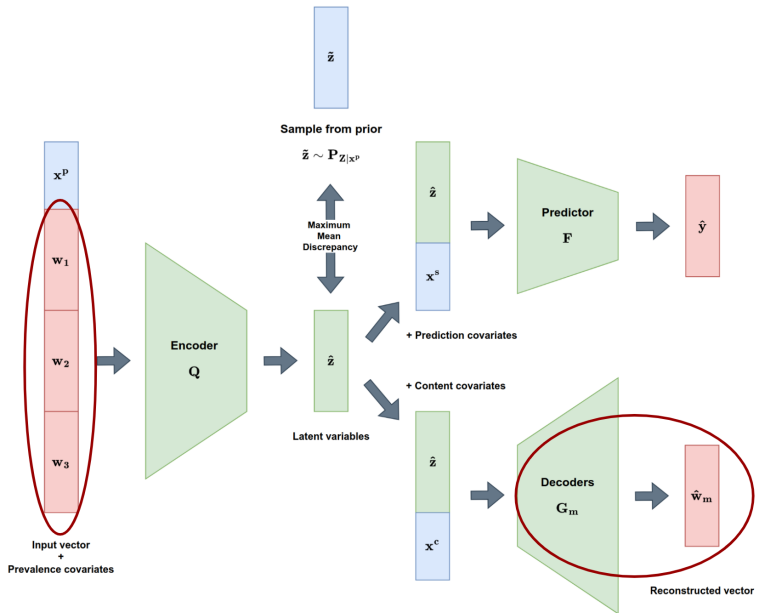
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Trading Off Prediction and Topic Quality

Approach	Pred Weight, λ_1	R^2 Train	R^2 Test	Topic Coherence
LDA+Lasso	-	0.030	0.023	0.74
GTM	0	-	-	0.464
	e^9	0.659	0.025	0.372
	e^{10}	0.699	0.051	0.244
	e^{11}	0.626	0.066	0.248
	e^{20}	0.321	0.076	0.098

- Increasing prediction weight increases test-set R^2 at cost of lower topic coherence (distinctiveness of top phrases).
- LDA gets worse prediction, but better topics, compared to GTM.



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Using Multilingual or Multimodal Data

- Aligned corpus approach (e.g. bag-of-words):
 - Need observations in the corpus with both modalities. If not available in original data, produce translations for (some of) the corpus.
 - Each modality (different languages or different media types) has its own encoder and own decoder.
 - Encoder is a "mixture of experts" where the cross-modal embeddings are multiplied together. Can assign topics to docs with only one modality.
- Multilingual/modal embedding approach:
 - Use a pre-trained multilingual embedding model that produces comparable embeddings across languages (or correspondingly, a multimodal embedding that produces comparable embeddings across media types).
 - Use those vectors as inputs to the GTM encoder. Should still decode as bag-of-words in the original language.
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 - Use those vectors as inputs to the GTM encoder. Should still decode as bag-of-words in the original language.
 - No translation needed. But requires data to be somewhat comparable across languages/modalities.

Multilingual Example

- Take 11K Wikipedia articles in Chinese and English from six pre-specified topics (food, religion, movies, music, politics, nature).
- Fit GTM with $k = 6$, using aligned-corpus approach (with translations) and multilingual pre-trained embeddings approach.

Input	Train Language(s)	Test Language(s)	% of Aligned Docs	Accuracy (%)
Bag of Words	English	English	-	91 (1.39)
	Chinese	Chinese	-	85 (2.15)
	English & Chinese	English & Chinese	100	85 (1.43)
			50	84 (0.85)
			10	80 (1.63)
Embeddings	English	English	0	76 (4.08)
	English	Chinese	0	72 (2.56)
	Chinese	Chinese	0	75 (2.97)
	Chinese	English	0	77 (1.54)

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Multimodality: Text/Images on Politician Facebook Groups

- We collect the universe of posts containing both text and images shared by US Congress representatives on Facebook, 2010-2020.
- Fit GTM on texts using LongCLIP embeddings, which handle either texts or images as input. Reconstruct the bag-of-words representation as output.
- Assign topics out-of-sample to both texts and images.

20 topics learned from Facebook posts

Label	Representative Words
Sports and Competitions	game, win, team, football, fun
Holidays and Awards	wish, family, love, present, hope
Voting System	vote, election, democracy, ballot, state
Military Honors	brave, medal, hero, military, present
Gun Violence	gun, violence, shooting, action, border
COVID-19 Health Services	vaccine, test, health, covid, site
Natural Disasters	storm, fire, damage, evacuation, area
Clean Energy & Infrastructure	energy, infrastructure, clean, water, project
Campaign Volunteering & Outreach	volunteer, campaign, update, constituent, folk
Tragedy & Mourning	lose, prayer, love, fight, attack
Women's Rights	woman, equality, vote, fight, legacy
Voting Encouragement	vote, early, polling, ballot, location
Taxes	tax, cost, insurance, bill, plan
Business Openings	facility, manufacturing, opening, company, economy
Public Health	health, funding, flood, fire, recovery
Military Sacrifice & Remembrance	sacrifice, hero, brave, forget, woman
Economy	economy, tax, growth, cost, american
Farming & Small Business	farmer, product, company, grow, industry
Schools	teacher, school, education, class, young
Immigration & Border Control	border, immigrant, illegal, crisis, southern

Some examples: Military Honors

Top Topic: Military Honors, Share: 0.05, Weight: 0.72



Some examples: Business Openings



Some examples: Schools

Top Topic: Education & Youth, Share: 0.11, Weight: 0.96



Some examples: Immigration and Border Controls

Top Topic: Immigration & Border Control, Share: 0.06, Weight: 0.90



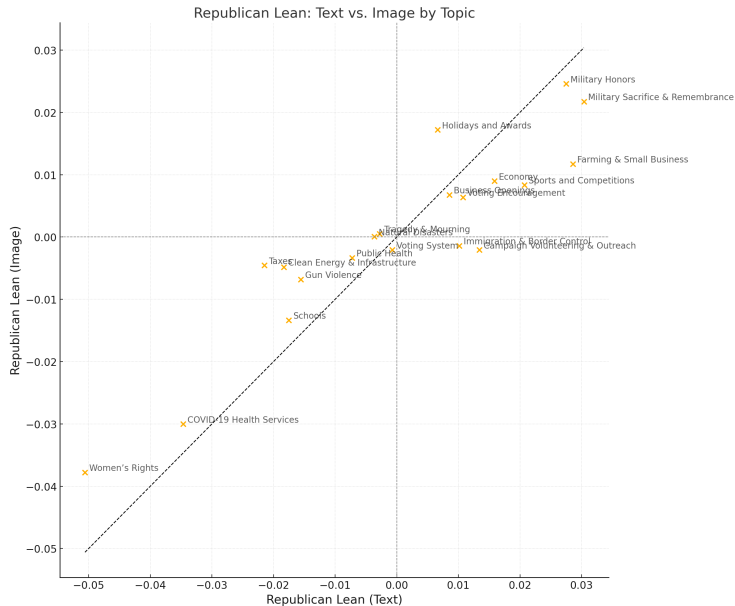
Which Topics are Expressed through Images?

Topic Code	Topic Label	Prevalence (Text)	Prevalence (Image)	Image-Text (%)
5	Gun Violence	0.045	0.081	80.30
2	Holidays and Awards	0.047	0.084	77.27
3	Voting System	0.023	0.039	71.94
10	Tragedy & Mourning	0.041	0.066	61.10
20	Immigration & Border Control	0.039	0.060	53.90
11	Women's Rights	0.051	0.074	45.10
12	Voting Encouragement	0.061	0.083	36.68
19	Schools	0.084	0.106	25.99
4	Military Honors	0.055	0.050	-9.45
9	Campaign Volunteering & Outreach	0.080	0.070	-12.20
6	COVID-19 Health Services	0.053	0.046	-12.76
7	Natural Disasters	0.051	0.044	-13.99
1	Sports and Competitions	0.060	0.051	-14.66
16	Military Sacrifice & Remembrance	0.041	0.034	-15.53
14	Business Openings	0.042	0.030	-30.04
17	Economy	0.050	0.022	-55.54
18	Farming & Small Business	0.046	0.017	-61.83
8	Clean Energy & Infrastructure	0.048	0.018	-62.06
13	Taxes	0.045	0.015	-65.42
15	Public Health	0.039	0.008	-79.01

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Image Slant Tracks Text Slant



Why (else) use GTM?

- Built on structural statistical model of documents
- Fast, modular, and scalable system
- Incorporates covariates that influence topic choice and framing
- Learns embeddings and can take pre-trained embeddings as input
- Powerful supervised learning method to predict outcomes
- Multilingual and multimodal

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GTM: User-friendly code base

github.com/PinchOfData/DeepLatent

```
# 2. Define modalities using fitted vectorizer
modalities = {
    "default": {
        "column": "doc_clean",
        "views": {
            "bow": {
                "type": "bow",
                "vectorizer": vectorizer
            }
        }
    }
}

# 3. Create GTMCorpus with prevalence and content covariates
train_dataset = GTMCorpus(
    df=df,
    modalities=modalities,
    prevalence=~ C(party) + C(chamber) + C(year)",
    content=~ C(party) + C(chamber) + C(year)"
)

# 4. Train the model
tm = GTM(
    train_dataset,
    n_topics=20,
    doc_topic_prior="dirichlet",
    update_prior=True,
    initialization=True,
    dropout=0.2,
    print_content_covariates = False,
    encoder_hidden_layers=[256],
    decoder_hidden_layers=[256],
    w_prior=1,
    patience=3
)
```

```
print(
    "\n".join(
        [
            "{}: {}".format(str(k), str(v))
            for k, v in tm.get_topic_words(topK=5).items()
        ]
    )
)
```

```
Topic_0: ['bosnian', 'sanction', 'unilateral', 'iranian', 'palestinian']
Topic_1: ['offender', 'semiautomatic', 'criminal', 'firearm', 'crime']
Topic_2: ['abortion', 'fetus', 'abortionist', 'unborn', 'embryonic']
Topic_3: ['reinsurance', 'reinsurer', 'hedging', 'insurer', 'insurance']
Topic_4: ['algal', 'bloom', 'nuclear', 'cell', 'otter']
Topic_5: ['wage', 'tax', 'minimum', 'wealthy', 'taxi']
Topic_6: ['championship', 'eulogy', 'primal', 'baseball', 'pitcher']
Topic_7: ['ordination', 'pastor', 'church', 'congregation', 'pastorate']
Topic_8: ['mental', 'pregnancy', 'dental', 'disorder', 'veteran']
Topic_9: ['tax', 'surplus', 'deficit', 'budget', 'spending']
Topic_10: ['court', 'defendant', 'clause', 'subpoena', 'firearm']
Topic_11: ['championship', 'coach', 'inning', 'coaching', 'officiate']
Topic_12: ['armenian', 'genocide', 'atrocities', 'massacre', 'extermination']
Topic_13: ['discretionary', 'earmark', 'readiness', 'budget', 'defense']
Topic_14: ['census', 'bureau', 'applicator', 'mammal', 'fishery']
Topic_15: ['trade', 'emission', 'oil', 'export', 'import']
Topic_16: ['recess', 'ripen', 'appropriation', 'recommit', 'tuesday']
Topic_17: ['appointment', 'coach', 'magistrate', 'brigadier', 'exemplary']
Topic_18: ['student', 'instructional', 'school', 'teacher', 'curriculum']
Topic_19: ['amnesty', 'voter', 'election', 'ballot', 'immigration']
```

Deep Latent Variable Models for Text and Audiovisual Data

Germain Gauthier, Philine Widmer, **Elliott Ash**

Machine Learning and Economics Summer Conference (MLESC)

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